Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 50%. We keep your highest score.

Next item →

1. Which component of the discriminator model is primarily responsible for feature extraction? 1 / 1 point

- ○ Fully connected layers
- ○ Pooling layers
- ○ Activation functions
- ◉ Convolutional layers

Correct
Correct! The convolutional layers are indeed responsible for feature extraction in the discriminator model.

2. What is the primary role of the generator in a Generative Adversarial Network (GAN)? 1 / 1 point

- ◉ To create realistic data samples from random noise.
- ○ To optimize the loss function for the GAN.
- ○ To compile and train the GAN model.
- ○ To discriminate between real and fake data samples.

Correct
Correct! The generator's role is to create realistic data samples that can trick the discriminator.

3. What are key steps involved in the compilation process of a GAN? 1 / 1 point

- [ ] Initializing weights to zero for faster training.
- [x] Compiling the models with appropriate loss functions and optimizers.

  Correct
  Correct! Compilation with the right loss functions and optimizers is crucial for effective training.

- [ ] Pre-training the discriminator before starting adversarial training.
- [x] Combining the generator and discriminator into a single model.

  Correct
  Correct! Combining the models is essential for training them adversarially.

- [x] Defining the architecture of both generator and discriminator models.

  Correct
  Correct! Defining the architecture for both models is the first step in the compilation process.

4. What are some considerations to keep in mind when training a DCGAN? 1 / 1 point

- [x] Monitoring for mode collapse

  Correct
  Correct! Mode collapse is a common issue that needs to be monitored.

- [x] Choosing the right optimizer

  Correct
  Correct! The optimizer plays a significant role in the training process.

- [ ] Using dropout layers extensively
- [x] Ensuring a balanced dataset

  Correct
  Correct! A balanced dataset is crucial for effective training.

- [ ] Regularly updating the activation functions

5. What is one primary challenge of training a Deep Convolutional Generative Adversarial Network (DCGAN) on a local computer? 1 / 1 point

- ◉ Limited computational resources
- ○ Lack of internet connectivity
- ○ Absence of a GPU
- ○ Inadequate storage space

Correct
Correct! Local computers often lack the necessary computational power to train complex models like DCGANs efficiently.